

SDM
FINANCIAL

Derivatives Guide Book

Understanding Options

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INTRODUCTION TO DERIVATIVES

Basic Definition

A derivative is a financial contract whose value is based on the performance of an underlying asset. The most common types of derivatives are options and futures.

- **Options** give the holder the right, but not the obligation, to buy or sell an underlying asset at a specific price and date.
- **Futures** are similar to options, but are agreements to buy or sell an underlying asset at a specific price and date.

Value Proposition

There are several reasons why someone might be interested in derivatives:

- **Hedging:** Derivatives may be used to hedge risks, such as the risk of fluctuating prices for commodities, currencies, or digital assets. For example, a farmer might use a futures contract to lock in a price for his crops, reducing the risk that he will sell them at a loss if prices drop.
- **Risk Management:** Large financial institutions use derivatives to manage the risk of their own portfolios, as well as the risk of their clients. This can help them reduce the overall level of risk in their core portfolio.
- **Generate Returns:** By selling derivatives, investors can collect premiums to generate additional revenue or yield. Additionally, investors can trade derivatives on a relative-value basis.
- **Limiting Downside:** Buying derivatives gives investors exposure to an asset class with the losses capped to the premium paid.
- **Speculation:** Investors may use derivatives as a way to speculate on future price movements of an underlying asset. For example, an investor may bet that a stock's price will go up within a specific time period by buying a call option, giving them the option to buy that stock at a given price.
- **Leverage:** Derivatives may be used to create leverage, meaning an investor can experience the financial impact of a large amount of an underlying asset with a relatively small amount of capital. This can increase the volatility of an investment, whether to the upside or downside.
- **Creating new Investment Opportunities:** Derivatives may be used to create new investment opportunities by trading in markets where it would be impractical to trade the underlying assets themselves.

WHY CRYPTOCURRENCY DERIVATIVES?

There are several risks and challenges that investors face with trading digital assets:

- **Price risk:** Digital assets are highly volatile and the price can fluctuate significantly over short periods of time. By not hedging these risks investors may be exposed to the risk of losses if the price of the digital asset moves against their position.

- **Limited flexibility:** Derivatives open the door to more specific risk management or portfolio management strategies that spot trading alone simply cannot match. Using derivatives allows an investor to limit losses, collect immediate returns in exchange for capping upside potential, or remain direction-agnostic while gaining exposure to market volatility in an underlying asset.

Trading derivatives in digital assets can be a valuable alternative for gaining exposure to digital assets without holding the underlying asset. It also gives an opportunity to hedge against risks, and speculate on future price movements, while potentially limiting downside exposure. Derivatives can be a healthy part of a balanced investment portfolio but must be treated with caution as a potential leveraging tool.

WHO CAN BENEFIT FROM TRADING CRYPTOCURRENCY DERIVATIVES?

Since derivative strategies may be tailored to an investor's requirements and risk profile, they may be used with success by a wide range of investors and across various use cases. Options specifically are a popular tool for deploying risk management and alpha trading strategies. The range of investors seen using derivatives in the digital asset industry include but are not limited to:

- Miners
- Institutional traders and hedge funds
- Treasurers and Corporations
- Family Offices and HNWIs
- Payment Processors and BTM Operators
- Blockchain Projects and Token Issuers

Here are some examples of how derivatives can be used:

Miners

A bitcoin mining facility may choose to buy bitcoin derivatives to:

- Hedge long and short positions in bitcoin from holdings:

  If a mining facility holds a large bitcoin position or expects to sell a specific number of bitcoins at a future date, miners may buy put options to limit potential losses should the price of bitcoin decline.

- Generate Alpha or Additional Income:

  Miners tend to have long positions in Bitcoin due to a mined inventory and anticipated mining rewards. Hence, can sell out-of-the-money calls to collect premiums on top of those long positions. If prices do not rise past the option strike price, the miner generates additional revenue through premium without being forced to sell away their bitcoin position. If the price rises past the strike price, the miner's long position profits from upward price movement in bitcoin, until the strike price, in addition to the premium. The miner's risk in this would be the opportunity cost of that additional income.

Hedge Funds and Institutional Traders

Hedge funds and institutional traders may use digital asset derivatives in a variety of ways, some of which are listed below:

- Hedge against downside risk of an underlying asset.

  Purchasing put options on an asset can protect against a potential price decline.

- Trade relative value across the volatility surface.

  It is possible for individuals or entities to generate profits through the trading of a diverse range of options across varying tenors and strikes, with the aim of capitalizing on any inefficiencies present in the volatility surface.

- Speculate on the price movement of the underlying asset.

  Purchasing call options on an underlying asset, betting that the price will increase in the future.

- Arbitrage by taking advantage of price discrepancies between different exchanges.

  Individuals or entities may sell a futures contract on one exchange and simultaneously buy the underlying asset on another exchange, profiting from the price difference.

- Gain leverage and amplify their returns.

  Margin trading may be used to trade a large position in an asset with a small amount of capital.

Family Offices and HNWIs

Here are some other reasons why Family Officers and HNWIs would be interested in derivatives:

- Diversification

  Investors can get direct exposure to digital assets without holding the underlying asset, as well as help in building a non-correlated asset portfolio

- Risk management

  Options contracts can be used to set a limit on the maximum potential loss on a digital asset position.

- Portfolio optimization

  Individuals or entities may use futures or options to take short positions on assets that are overvalued and long positions on assets that are undervalued.

Payment Processors, BTM Operators, Blockchain Projects and Token Issuers

Main use case for this group would be:

- Hedging against price volatility

  Options contracts can be used to hedge against the risk of price fluctuations and risk associated with holding large amounts of digital assets.



OPTIONS PRIMER

Introduction to Options

An option is a financial contract that gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a specified price (strike price) on or before a specified date (expiration date). There are two main types of options: call options and put options.

- A call option gives the holder the right to, but not the obligation, to buy an underlying asset at a specified price.
- A put option gives the holder the right, but not the obligation, to sell an underlying asset at a specified price.

If the holder chooses to exercise their option, the seller of the option (or writer) is obligated to buy or sell the underlying asset at the strike price. If the holder chooses not to exercise their option, the option expires worthless. Investors pay a fee, referred to as the options premium, to purchase an options contract. The fee represents the price that the options writer charges. It is determined by various factors such as the price of the underlying asset, strike price, time remaining until expiration, volatility of the underlying asset, and the risk-free rate.

There are two types of options based on how they are exercised:

- American Options: Can be exercised anytime before the option expiration date.

- European Options: Can be exercised on the day of the option expiration.

Terminology Recap

Call Option: Type of option contract that allows the holder the choice to buy the underlying asset from the writer.

Put Option: Type of option contract that allows the holder the choice to sell the underlying asset to the writer.

Strike Price: Price at which the option contract allows the holder to buy or sell the underlying asset.

Expiration date: Last date that the holder can exercise the option contract. Holders can exercise it at any time before if it's an American option, and can only be exercised on the day of its expiration if it's a European option.

Options Payoff Diagram

A payoff diagram is a graphical representation of the potential profit or loss of an option position at expiration. It shows the relationship between the price of the underlying asset and the profit or loss of the option position.

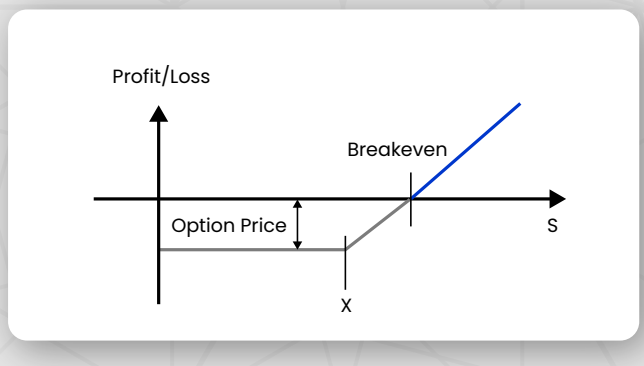
In these diagrams, the vertical axis represents the profit/loss or the payoff, and the horizontal axis (S) represents the price of the underlying asset. X represents the strike price. Payoff diagrams are useful for visualizing the potential outcomes of an option position and for comparing different option strategies.

The basic option positions with their respective payoff diagrams are as follows:

Option Positions

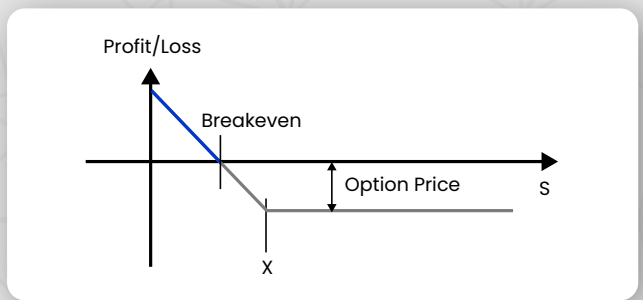
There are two parties involved in an options contract, the buyer (or holder) going "long", and the seller (or writer) going "short". Additionally, there are two types of options: calls and puts. Therefore, there are four basic option positions:

- Long Call



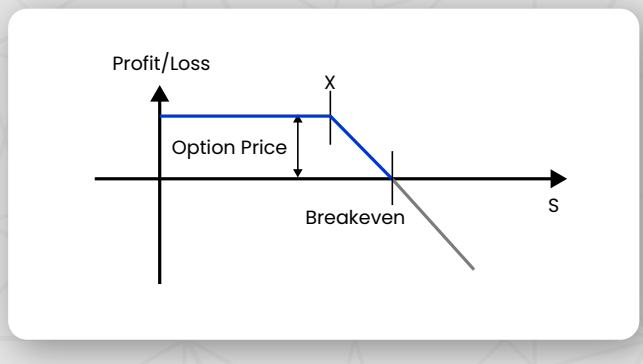
The holder of a call option pays a premium to be able to buy the assets at a predefined strike price. If the underlying asset falls below the strike price, their loss is limited to the premium paid. On the other hand, holders benefit when the price of the underlying asset increases above the strike price. The holder would exercise the option to buy the underlying asset, profiting the difference from the higher spot price and the strike price.

- Long Put



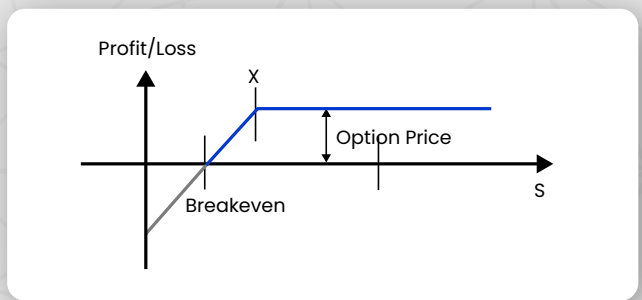
The holder of a put option pays a premium to be able to sell the assets at a predefined strike price. If the underlying asset rises above the strike price, its loss is limited to the premium paid. However, holders benefit when the underlying asset decreases below the strike price. Hence would exercise the option to sell the underlying asset, profiting the difference from the lower spot price and the strike price

- Short Call



The seller of a call option collects a premium from the buyer of the call option allowing the call holder to buy at a predefined strike price. If the underlying asset rises above the strike price, its loss is the difference between the spot and the strike price, with the potential for unlimited downside. However, sellers benefit when the price of the underlying asset stays below the strike price, allowing them to keep the option premium without having to sell the underlying asset.

- Short Put



The seller of a put option collects a premium from the buyer of the put option allowing the put holder to sell at a predefined strike price. If the underlying asset falls below the strike price, their loss is the difference between the spot and the strike price, with a downside equivalent to the strike price less premium. Sellers benefit when the price of the underlying asset stays above the strike price, allowing them to keep the option premium without having to sell the underlying asset.
When buying a long position i.e. holding a call or put option, the holders risk is limited to only the premium spent to hold those contracts. On the other hand, when writing a short position i.e. selling a call or put option, sellers risk is dependent on the price difference between the underlying asset and the strike price. This implies that the option sellers have exposure to more, and in some cases, unlimited risks.
Here is a table summarizing the above:

Option Position	Maximum Gain	Maximum Loss
Long Call	**Unlimited**	Premium
Long Put	Limited to strike price less premium	Premium
Short Call	**Premium**	Unlimited
Short Put	**Premium**	Strike price less premium

Option Value

Option value is based on the following:

- Intrinsic Value

  The intrinsic value of an option is the difference between the current price of the underlying asset and the strike price. The intrinsic value defines the “moneyness” of options. The option is in-the-money (ITM) when the intrinsic value is positive and is out-the-money (OTM) when the intrinsic value is negative.

  It is at-the-money (ATM) when the intrinsic value is zero, meaning the price of the asset equals the strike price

- Time Value

  The time value of an option, also known as the "extrinsic value", represents the value that an option has beyond its intrinsic value. It takes into account the possibility of the underlying asset price moving in the holder's favor before the option expires, as well as other factors such as volatility and interest rates.

  The time value of an option decreases as the option gets closer to expiration. This is because as the option gets closer to expiration, there is less time for the underlying asset price to move in the holder's favor and the intrinsic value of the option becomes more certain.

Factors that affect Option Pricing

Here are some of the other factors that affect option premiums.

- **Time to expiration**

  The time remaining until the option's expiration date is known as time to expiration. As expiration approaches, time decay (or theta) of the option increases, which means the option premium decreases, all else being equal.

- **Implied volatility**

  Implied volatility reflects the market's expectation of the future volatility of the underlying asset. Options on assets with higher implied volatility will have a higher option premium than options on assets with lower implied volatility.

- **Interest rates**

  The interest rate environment can affect the price of an option because it affects the cost of holding an option position. A higher interest rate will increase the cost of holding a long option position and decrease the cost of holding a short option position.

- **Dividend**

  Dividends paid by the underlying asset can also affect the price of an option. A dividend payment can reduce the price of the underlying asset and thus decrease the intrinsic value of call options while increasing the intrinsic value of put options.

- **Supply and demand**

  The supply and demand of options on the market also affect the option premium. An increase in demand for options will increase the premium, while an increase in supply will decrease the premium.

These are the main factors that affect the price of an option. These factors are interrelated and affect the option premium in different ways. Understanding how these factors work together is important for pricing options and making informed trading decisions.

Option Chain

An options chain, also known as an option matrix, is a listing of all available options contracts for a given security. The chain shows all listed puts and calls, as well as their expiration, strike prices, volume, and pricing information for a single underlying asset within a given maturity period. Below is the screenshot of the call option chain for BTC.

26 JAN 23 (17h) (Daily)					Calls		26 JAN 23
Size IV (Bid)	Bid	Mark	Ask	IV (Ask)	Size	AIDelta Position	Strike
31.7	0.1985 $4478.09	0.2021 74.53%	0.2110 $4760.08	414.2%	32.9	1.00	18000
32.4	0.1540 $3474.18	0.1578 74.53%	0.1665 $3756.18	336.7%	33.1	1.00	19000
34.8	0.1085 $2448.78	0.1138 74.53%	0.1210 $2730.90	246.5%	33.7	1.00	20000
5.5	0.0875 $1974.82	0.0917 74.53%	0.1000 $2256.94	221.7%	31.8	1.00	20500
4.5	0.0660 $1489 58	0.0697 72.77%	0.0785 $1771.70	189.5%	25.6	0 99	21000

An option chain typically includes the following columns:

- **Strike price**

  This column displays the strike price of the options available. The strike price is the fixed price at which the underlying asset can be bought or sold if the option is exercised.

- **Expiration date**

  This displays the expiration date of the options available. The expiration date is the date after which the option can no longer be exercised.

- **Type**

  This column displays the type of option, either call or put.

- **Bid price**

  This column displays the bid price of the option, which is the highest price that a buyer is willing to pay for the option.

- **Ask price**

  This column displays the ask price of the option, which is the lowest price that a seller is willing to accept for the option.

- **Last price**

  This column displays the last traded price of the option, which is the price at which the option was last traded.

- **Open interest**

  This column displays the open interest of the option, which is the number of outstanding options contracts that have been bought but not yet sold or exercised.

- **Volume**

  This column displays the volume of the option, which is the number of options contracts that have been traded during a certain period of time.



Options Risks: Greeks

Many of the risks associated with options can also be modeled and understood. This particular feature of options actually makes them arguably less risky than other asset classes, or at least allows the risks associated with options to be understood and evaluated. Individual risks have been assigned Greek letter names, and are sometimes referred to simply as "the Greeks."
Greeks in option pricing are a set of parameters used to measure the sensitivity of the price of an option to various factors, such as the underlying asset's volatility or time to expiration. They provide insight into how different changes may affect the value of the option.

The basic Greeks include:

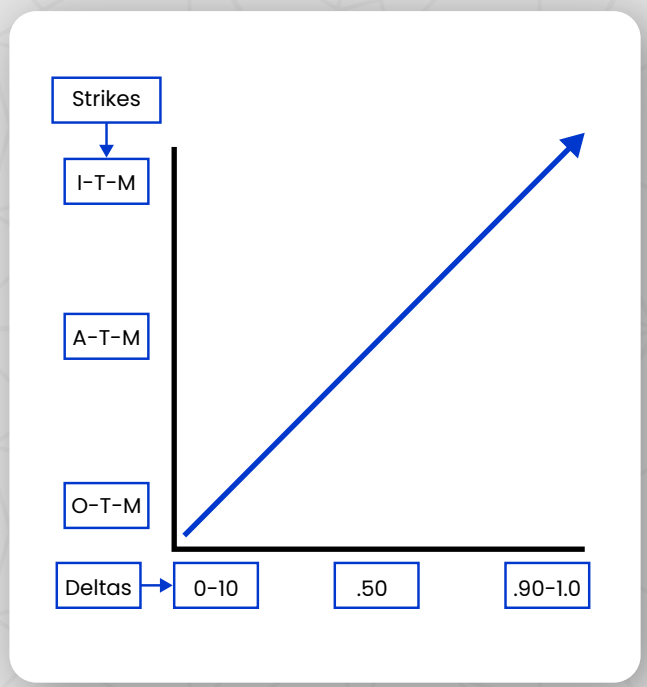
- Delta: option's price sensitivity to price changes in the underlying asset
- Gamma: option's delta sensitivity to price changes in the underlying asset
- Theta: time decay, or option's price sensitivity to the passage of time
- Vega: option's price sensitivity to changes in volatility
- Rho: option's price sensitivity to interest rate changes

Delta (Δ)

elta refers to the amount an option price is expected to change when the underlying asset moves by one point. Delta is used as a measure in trading to determine how much money will be put at risk when taking a position on options.

For example, if a call option has a delta of 0.5, it means that if the underlying asset price increases by $1, the call option's price will increase by $0.50.

The delta of a call option will typically be a positive number, since an increase in the price of the underlying asset increases the value of a call option. For put options, delta is generally negative since an increase in the value of the underlying asset decreases the value of a put option.



As an option gets further in-the-money, the probability it will be in-the-money at expiration increases as well. So the option's delta will increase. As a general rule, in-the-money option prices will move more than out-of-the-money options, and short-term options will react more than longer-term options to the same price change in the asset. Additionally, ATM options will have a delta close to .5, as the option has roughly a 50% chance to finish in-the-money.

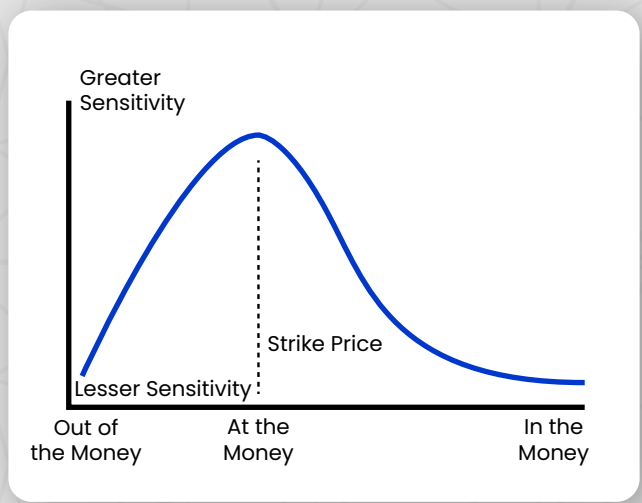
As expiration approaches, changes in the asset's value will cause more dramatic changes in delta due to increased or decreased probability of finishing in-the-money.

Gamma (Γ)

Gamma is a measure of the rate of change in an option's delta. If delta is the "speed" at which option prices change, you can think of gamma as the "acceleration." This makes Delta the rate of change in an option's price given a one-point move in the underlying.

For example, if a call option has a gamma of 0.02, it means that if the underlying asset price increases by $1, the call option's delta will increase by 0.02.
Gamma measures how much a delta changes for every one-point move in the underlying asset's price, and thus it helps quantify how much risk and profit potential an option has for its holder prior to expiration. Gamma may be used by traders to gauge whether the position held has been appropriately hedged against changes in market conditions, and how much additional hedging may be required should those conditions shift again.

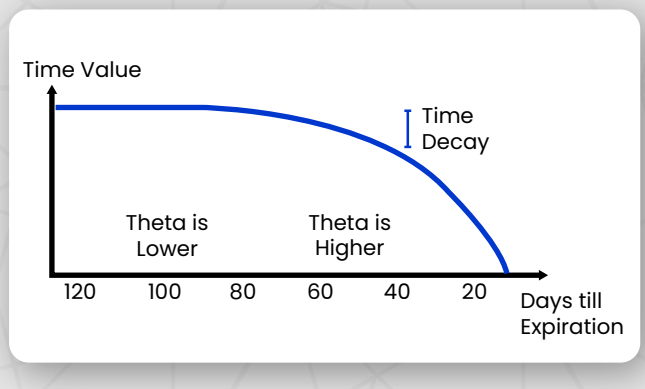
The higher the gamma, the more likely it is that delta will need to be adjusted as market prices fluctuate and expiration approaches.



Gamma increases as an option nears the at-the-money (ATM) point and declines the further out-of-the-money (OTM) or in-the-money (ITM) it is. At-the-money options have the highest gamma, because their deltas are the most sensitive to underlying price changes. Moreover, very short-dated ATM options have a very high gamma.

Theta (Ø)

Theta is a measure of time decay in option prices. It reflects the rate of change in an option's price over a period of time due to the passage of time.
For example, if a call option has a theta of -0.05, it means that if one day passes, the call option's price will decrease by $0.05.



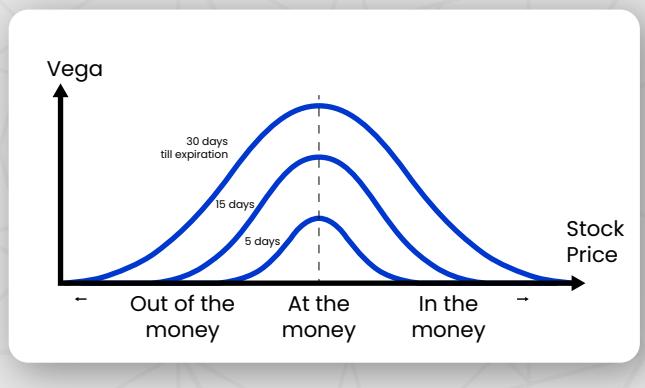
Higher Theta is an indication that the value of the option will decay more rapidly over time. Theta is typically higher for short-dated options, especially near-the-money, as there is more urgency for the underlying to move in the money before expiration. Theta is negative for long option positions because they lose value the longer they are held due to the erosion of time value and the increased likelihood that they will expire worthless.

At-the-money options will experience more significant losses over time than in- or out-of-the-money options with the same underlying and expiration date. This is because at-the-money options have the most time value built into the premium.

Vega (V)

Vega is the measure of the sensitivity of an option's price to changes in implied volatility. It is calculated by taking the partial derivative of the option's price with respect to implied volatility. It gives investors insight into how much buying or selling power a change in volatility can give them.

For example, if a call option has a vega of 0.10, it means that if the volatility of the underlying asset increases by 1%, the call option's price will increase by $0.10.



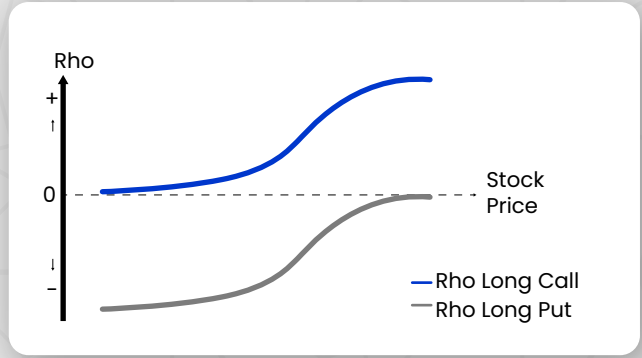
The longer an option contract has until it expires, the more volatility affects the price. Vega falls as the option gets closer to expiration and increases as the underlying moves closer to the strike. In other words, Vega is centered around at-the-money options and falls as it moves out-of-the-money or in-the-money.

Longer-dated options have a higher Vega value. This is a reflection of Vega's sensitivity to time. The more time an option has to be above or below its strike, the more sensitive the option will be to changes in implied volatility. Vega does not affect the intrinsic value of an option's premium, only the extrinsic value.

Rho (ρ)

Rho is a measure of the sensitivity of an option's price to changes in the interest rate. It measures how much the option premium will change when there is a one-percentage-point change in the interest rate.

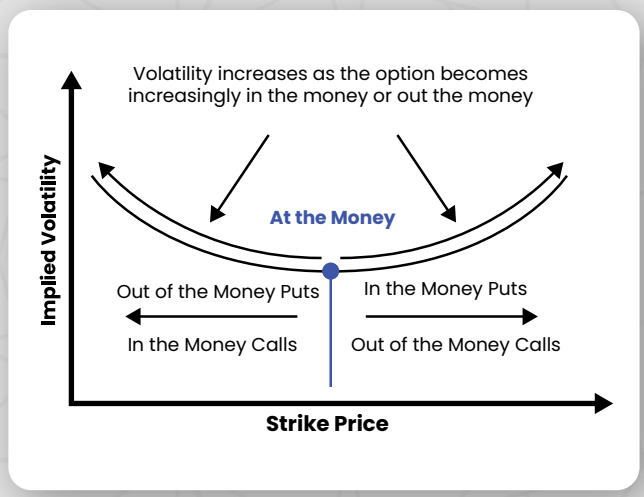
For example, if a call option has a rho of 0.01, it means that if the interest rate increases by 1%, the call option's price will increase by $0.01.



Rho can be positive or negative, depending on whether an increase in interest rates benefits the holder of the option. Rho is positive for long calls (right to buy) and increases with the price of the asset. Rho is negative for long puts (right to sell) and approaches zero as the asset price increases. Rho is positive for short puts (obligation to buy) and negative for short calls (obligation to sell).

Volatility Skew and Smile

Volatility skew is an options trading concept that refers to the difference in volatility between at-the-money options, in-the-money options, and out-of-the-money options. Volatility skew describes the observation that not all options on the same underlying and expiration have the same implied volatility assigned to them in the market. It's a way of comparing the differences in the prices of options with different strike prices. A positive skew means that options with higher strike prices have higher implied volatilities, while a negative skew means that options with lower strike prices have higher implied volatilities.



Volatility smile refers to the shape of the volatility skew, which is typically upward-sloping, creating a smile-like shape on a graph. The smile shape is due to the fact that options with strike prices away from the current asset price are more sensitive to changes in asset price, and therefore have higher implied volatilities.

Option Pricing

Option pricing is a complex and multi-disciplinary field that requires a good understanding of financial mathematics, statistical theory, and market dynamics. It depends on several factors including the price of the underlying asset, the strike price, the time to expiration, the volatility of the underlying asset, and the risk-free rate of return.

The value of the options is based on the sum total of its:

1. Intrinsic value
2. Extrinsic value

Intrinsic Value

The intrinsic value of an option is the inherent value of the option that is derived from the underlying asset. If the option is in-the-money, it is the difference between the current market price of the underlying asset and the strike price of the option. For at-the-money options or out-of-the-money options there is no intrinsic value.

For example, suppose the market price of Bitcoin is $25,000 and the strike price of the call option is $23,000. Then the intrinsic value of the call option is $2,000. If the price of Bitcoin falls to $20,000. There is no intrinsic value. It is important to note that the intrinsic value of an option represents the minimum value of the option and is not a guarantee of its value.

Extrinsic Value

The value of an option also includes an extrinsic component, which is known as the time value. It reflects the uncertainty of future market movements and the time remaining until the option's expiration. The longer the time until expiration, the greater the uncertainty and the higher the extrinsic value of the option. As the expiration date approaches, the extrinsic value decreases as the uncertainty decreases. For at-the-money options or out-of-the-money options, the entire value of the option is extrinsic, as there is no intrinsic value. Their value is derived from the possibility that they may become in-the-money before expiration.

There are several pricing models that have been developed to calculate the price of an option. These models differ in their complexity and the level of accuracy they provide. Some of the most widely used pricing models are:

1. Black-Scholes Model: This is one of the most widely used option pricing models. It uses five variables to determine the price of an option: the price of the underlying asset, the strike price, the time to expiration, the volatility of the underlying asset, and the risk-free rate of return. The model assumes that the underlying asset follows a log-normal distribution and that the holder can sell the option at any time prior to expiration.

2. Binomial Model: This model is an alternative to the Black-Scholes Model and is particularly useful for pricing options with early exercise features, such as American options. The model uses a tree structure to represent the possible price movements of the underlying asset and calculates the value of the option at each node.

3. Monte Carlo Simulation: This is a numerical method for option pricing that uses random numbers to simulate the movements of the underlying asset. The simulation is repeated many times to obtain an average value for the option. The model is particularly useful for pricing complex options, such as options with multiple underlying assets.

4. Implied Volatility Model: This model uses market data, such as option prices, to calculate the implied volatility of the underlying asset. The implied volatility is then used in one of the other pricing models, such as the SABR model, to calculate the price of the option.



Option Premium and Margin Requirements

In order to trade options, margin requirements must be met by traders before executing any trades. Margin requirements are based on the risk associated with each trade and can vary from dealer to dealer. When trading options, traders generally need to have cash or cash equivalents that meet or exceed a certain percentage of their total investment size as well as a certain level of equity set aside in case their positions move against them. This equity requirement helps to cover potential losses so that traders do not get into financial difficulty in case the markets move against them.

It's important to note that margin requirements can vary depending on the brokerage or exchange, and may also change depending on market conditions. It's always a good idea to check with our team for the most up-to-date margin requirements.

Long Options

For long options, call or put, the only requirement is the premium to be paid upfront. No additional margin is ever needed.

Short Options

For short options, if the position is covered, no additional margin is required as the collateral is being held on account.

Margin is usually 100% of the premium amount plus a minimum % requirement of the underlying notional.

Let's say a call option on Bitcoin with a strike price of $20,000 and a premium of $800 per coin is written or bought. The option is set to expire in 3 months. If the brokerage or exchange has a margin requirement of 20% for this option, the total margin requirement would be:

($20,000 (strike price) x 20%) + $800 (premium) = $4,800

So in this case, at least $4,800 would be needed in the account to buy this call option.

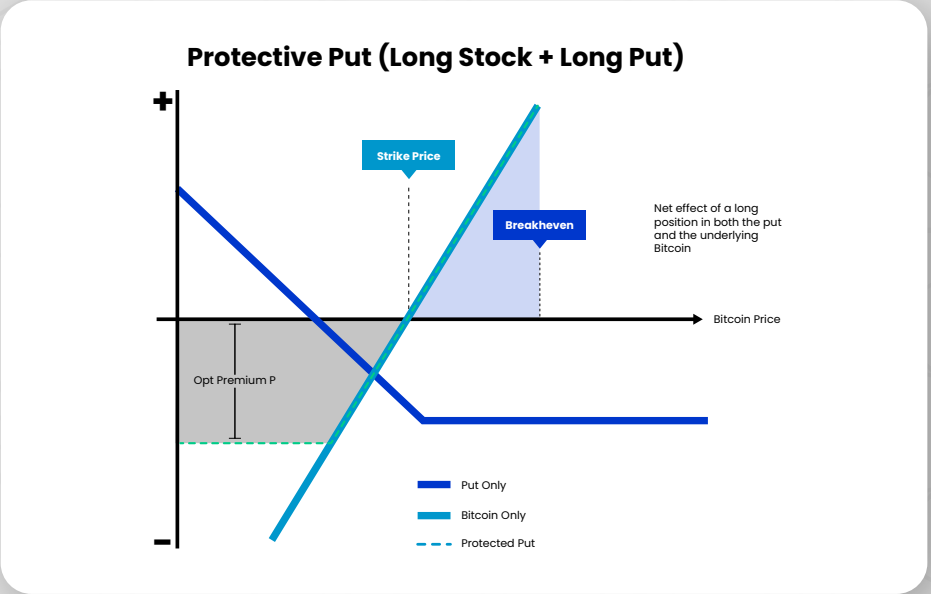
Options Strategies

Protective Put

Description

A protective put involves the simultaneous purchase of a put option and the underlying asset.

Payoff chart



Application

Protective puts are often used by investors to protect their existing long positions in a security against potential price declines. The investor pays a premium for the option, which gives them the right to sell their stock at a predetermined price if the market falls below that price. It essentially establishes a floor price at which you can sell your assets but the break-even price is increased which lowers the probability of profit. This technique can be useful to limit losses in markets that are volatile or bearish without having to liquidate the position entirely.

Investor Expectations

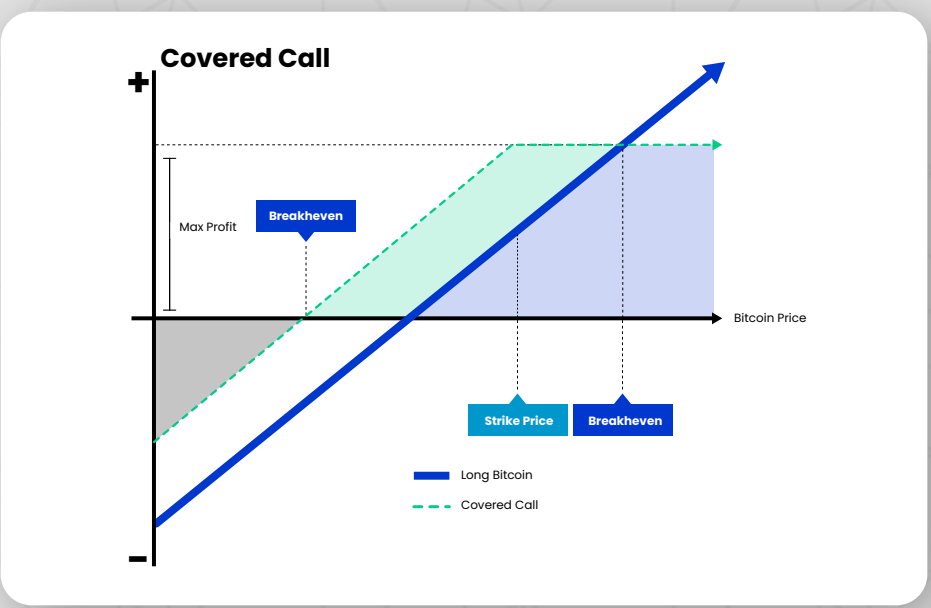
- Price Outlook: Bullish
- Volatility Outlook: Stable to Higher
- Potential Profit: Unlimited
- Risk: Limited

Covered Call

Description

A covered call involves the simultaneous writing of a call option and holding the underlying asset.

Payoff chart



Application

The strategy is advantageous when investors want to earn income on an underlying asset already owned, and are looking to hedge their portfolio, or expect limited upside potential from the underlying security. Covered calls through the sale of the call option generate income in the form of premium, but also limit the upside potential of the underlying asset position, as the investor is obligated to sell the asset at the strike price of the call option if the option is exercised. This strategy is considered "covered" because the investor has the underlying asset to deliver if the option is exercised.

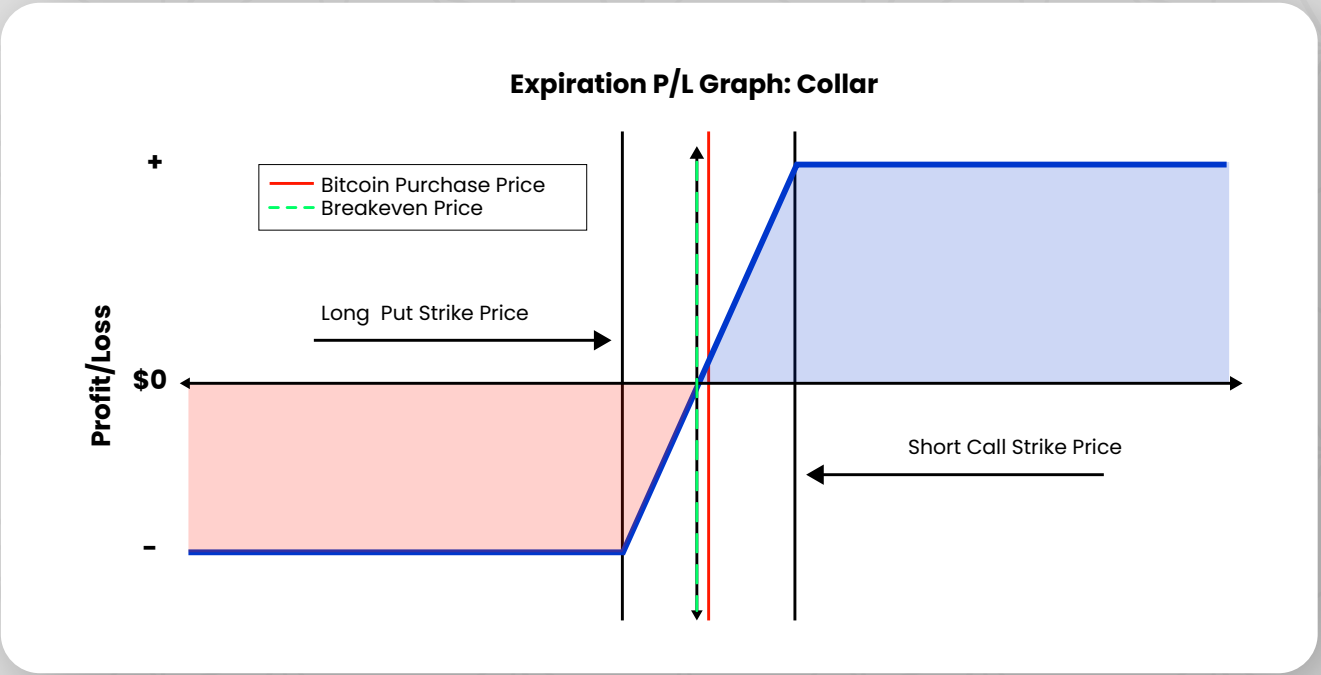
Investor Expectations

- Price Outlook: Moderately Bullish
- Volatility Outlook: Low
- Potential Profit: Limited
- Risk: Unlimited (max loss is strike price in actuality)

Collar (zero-cost) options

Description

A collar option is an options strategy in which an investor holds a position in an underlying while simultaneously purchasing protective puts and writing covered calls against the underlying to limit the downside risk. The investor sells an out-of-the-money call option and uses the proceeds to buy a protective put at the same time, thus forming a “collar” around the underlying.



Application

It is a common strategy to hedge at a very minimal cost. It protects investors from suffering large losses in a down move but limits the upside gains as well though. Only if the asset moves beyond either side of the “collar” (the strike price of the purchased and sold option) the losses and gains are limited from the option hedge.

Investor Expectations

- Price Outlook: Bullish
- Volatility Outlook: Medium
- Potential Profit: Limited
- Risk: Limited

EXAMPLE OPTION TRADES

Here are some real-world bitcoin option examples (same would apply for ETH options), broken down by:

a. Hedging downside price movement)
b. Positioning for a directional movement (bullish or bearish)
c. Generating alpha / additional income.

***for the examples below, current BTC market price = $24,000**

A. Hedging Against Downside Price Movement

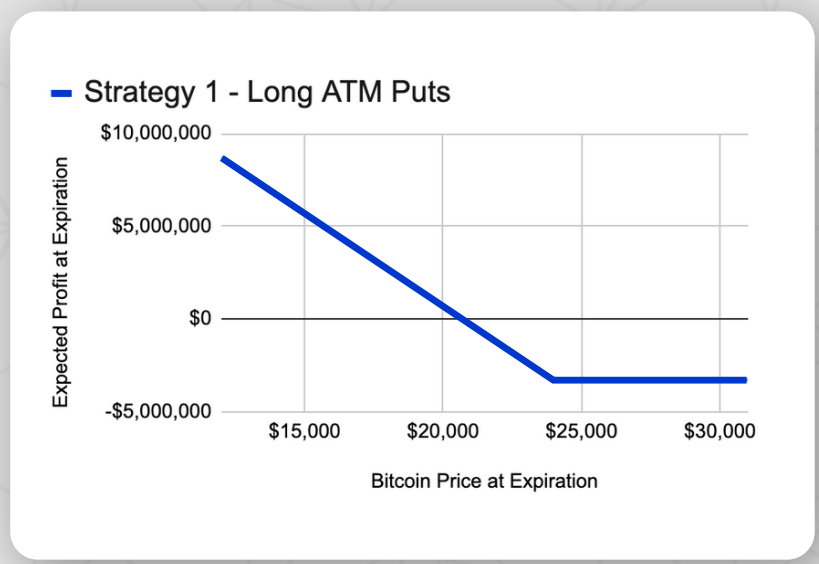
For those investors that are long the underlying token and would like downside protection, here are two examples:

1. Buy an at-the-money (ATM) put option, and
2. Buy a collar (buy put / sell call)

Example 1): Buy an ATM put option

Trade Details	Notional	Premium (BTC)	Premium (USD)	Comments
Buy June 30th expiry 24k strike BTC puts	1,000 BTC	.137	-$3,288,000	• Protects against downside movement • Effectively buying insurance against adverse movement • Can roll the strikes once option expiries, or unwind the option if the underlying position gets sold.

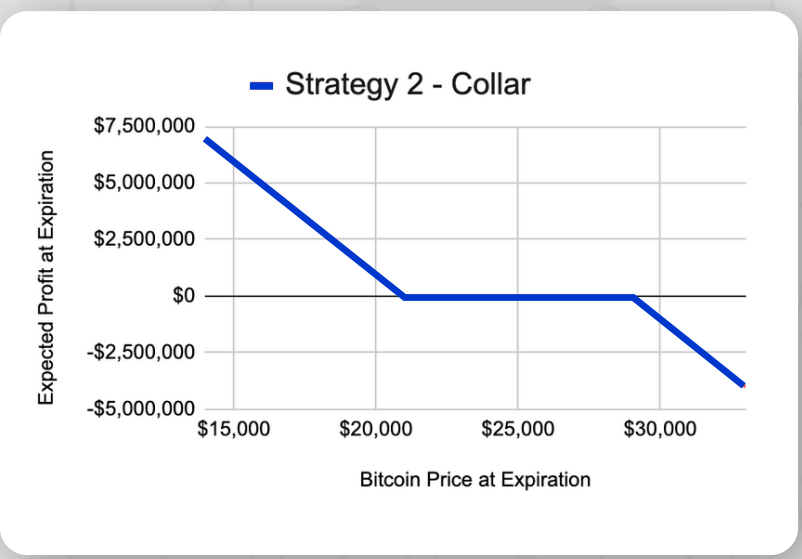
In this example, the investor is looking to hedge 1,000 BTC against BTC declining in value. Investors would purchase a put option with a 24k strike price which expires on June 30th. The investor is protected against all downside price action while knowing the maximum loss would be the premium paid. The upside still remains intact. The chart below shows the PNL (profit and loss) profile of the option.



Example 2): Buy a Collar (buy an OTM put vs selling OTM call)

Trade Details	Notional	Premium (BTC)	Premium (USD)	Comments
- Sell June 30th expiry 29k strike BTC calls - Buy June 30th expiry 21k strike BTC puts	1,000 BTC 1,000 BTC	.077 .078	+$1,848,000 -$1,872,000 Total: $24,000	• Protects against downside movement, but gives away some of the upside • Ideally would want the market to rally to just shy of the 29k strike • Low-cost way of protecting against adverse price movements

In this example, the investor is buying downside protection by entering into a long OTM put option position, but selling away some upside by selling the OTM call option against it. Specifically, for an investor long the underlying, he/she buys downside protection but gives up some of the upside after a certain point. In this example specifically, BTC is trading at 24k, and the 21k put option is purchased and mostly financed by selling the 29k call option. The investor would be protected on the downside if BTC moves below 21k, and still have upsides up to 29k. The chart below shows the PNL profile of the option.



B) Directional Positioning

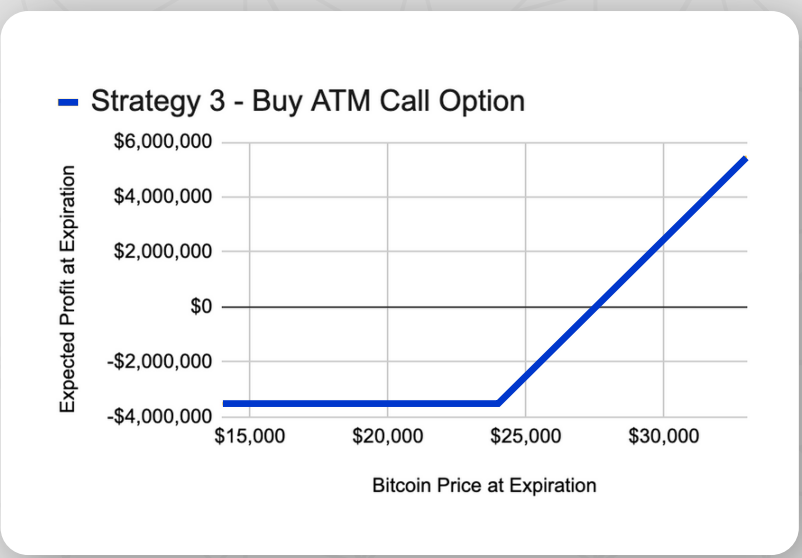
Investors who want directional exposure on a leveraged basis and/or with limited downside risk, the following are some examples:

1. Buy an ATM call option, and
2. Buying a call spread

Example 3): Buy an ATM Call Option

Trade Details	Notional	Premium (BTC)	Premium (USD)	Comments
Buy June 30th expiry 24k strike BTC calls	1,000 BTC	147	-$3,528,000	• Participate fully in the upside price movement • Limited downside (the premium paid) • Levered bullish strategy

In this example, the investor is looking for a bullish strategy for a way to get long BTC in a leveraged fashion. Investors would purchase a call option with a 24k strike price which expires on June 30th. The investor's loss is capped at the premium paid which is $3,528,000, but has upside on $24,000,000 of BTC. This is a very straightforward way to go long the market with limited downside. The chart below shows the PNL profile of the option.



Example 4) Buy a Call Spread

Trade Details	Notional	Premium (BTC)	Premium (USD)	Comments
- Buy June 30th expiry 24k strike BTC calls - Sell June 30th expiry 28k strike BTC calls	1,000 BTC 1,000 BTC	.147 .087	-$3,528,000 +$2,088,000 Total: $1,440,000	• Participate in the upside price movement from 24k to 28k • Limited downside to the option premium paid • Strategy is best implemented when the view is for a modest bullish rally

In this example, the investor is buying upside price action by buying ATM calls, but cheapening up the strategy by selling OTM calls against it. Specifically, investors will participate in a rally from 24k all the way to 28k. The most the investor can lose is the premium paid, which in this case is $1,440,000. The maximum profit an investor can realize with this strategy is $4,000,000, less the premium paid. The chart below shows the PNL profile of the option.



C) Alpha Generation

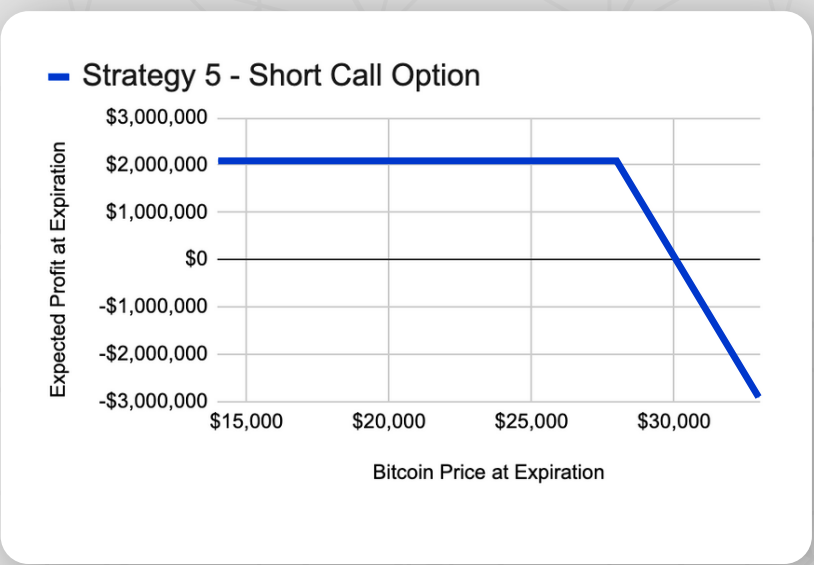
Investors looking to generate extra income by selling vol, here are two examples:

1) covered call, and
2) short iron condor.

Example 5) Buy Covered Call (Sell Call Option against long underlying)

Trade Details	Notional	Premium (BTC)	Premium (USD)	Comments
Sell June 30th expiry 28k strike BTC calls	1,000 BTC	087	-$2,088,000	• Participate fully in the upside price movement as investor is long the underlying asset • Generates additional alpha as the seller of the option collects the premium • Can roll the short calls on a regular basis to continue to collect premiums

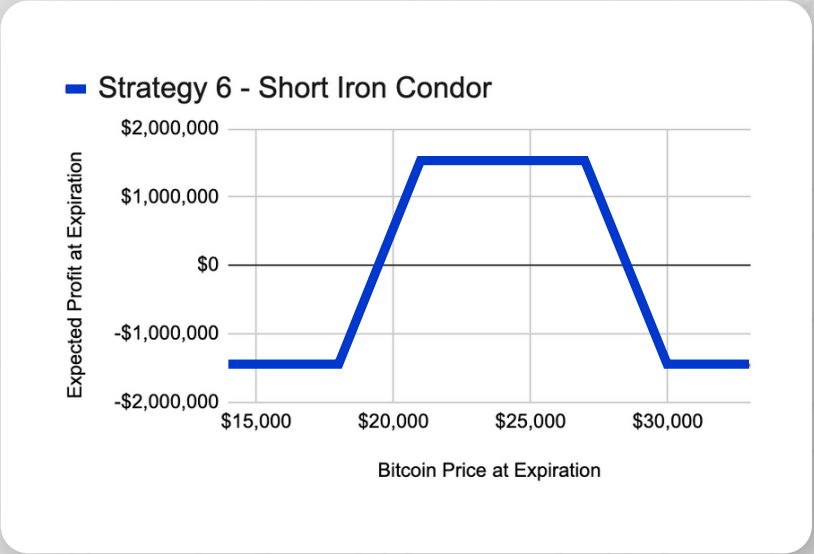
The investor is long the underlying asset but wants to generate additional alpha by selling away some of the upside over a certain period. The investor sells an OTM call against the long position in the underlying and collects the premium, hence generating additional returns on his long underlying position. In the event the market rallies past the 28k strike, the investor would make the 4k upside move (24k to 28k) plus the premium of $2,088,000, but would give up any additional upside. In the event the market stays here or goes lower, the $2,088,000 premium collected would act as a buffer against any losses on the underlying. The chart below shows the PNL profile of the option.



Example 6) Short Iron Condor

Trade Details	Notional	Premium (BTC)	Premium (USD)	Comments
- Short June 30th expiry 21k strike BTC puts	1,000 BTC	.0755	+$1,812,000	• Strategy involves selling a put spread and selling a call spread • Maximum upside is the premium collected • Maximum downside is the difference in the spread strikes (3k) less the premium collected, which equals $1,464,000
- Buy June 30th expiry 18k strike BTC puts	1,000 BTC	.04	-$960,000	
- Short June 30th expiry 27k strike calls	1,000 BTC	.099	+$2,376,000	
- Buy June 30th expiry 30k strike calls	1,000 BTC	.0705	$-1,692,000 Total: +$1,536,000	

This strategy is for an investor who believes the price of the asset will stay within a predefined range for a set period of time. Investors benefit by selling volatility and collecting the premium. The nice part about this strategy is that the downside is limited, versus a strategy like selling a straddle where downside is unlimited. If the market stays where it is or within a predefined range, in this case between approximately 20-28k, the investor would make money.



MOVING FORWARD

Here are a few questions to consider before executing a derivative trade:

What is the purpose of the trade?

- Is it to hedge an existing position, speculate on future market movements, or generate income?

What is my risk tolerance?

- Derivatives can be highly leveraged and carry significant risk, so it's important to understand the risk tolerance and how it fits in the overall risk management strategy.

What are the counterparty risks?

- Know how creditworthy the other party in the trade is, and put measures in place to mitigate counterparty risk.

What are the potential costs and fees associated with the trade?

- Make sure you understand any commissions, spreads, and other costs that may be associated with the trade.

What is the expiration date?

- Make sure you are aware of the expiration date of the contract and how it may affect the price.

What are my cash flow projections?

- You need enough liquidity to either pay the premium upfront and/or meet margin calls in order to keep the option trade openness. Plus the liquidity for potential margin and an understanding of your potential losses.

Have you done your research and analysis?

- Make sure you have done your research and analysis and know how the trade will be affected by changes in the underlying assets, volatility, interest rates, and other market factors.

What are the exit strategies?

- Make sure you know how the trade will be closed out, and what the potential consequences are of different exit scenarios. Make sure you have set your stop loss and take profit levels to manage your risk.

Understanding Legal Documents: ISDA and CSA

ISDA Agreement

The ISDA Master Agreement (as published by the International Swaps and Derivatives Association, the "Agreement") establishes a standardized contractual framework that greatly enhances and hastens the ability for two counterparties to engage in OTC derivative transactions, such as interest rate, currency, credit default swaps, and digital assets. It manages to do so by integrating several related, industry-accepted components, which are better understood by analogizing them as the 4 keys to the open road:

1. **The "Highway"** – the ISDA Pre-Print Form – This standardized portion of the agreement establishes the legal framework that envelopes each and every trade. It's best to think of it as a Highway – infrastructure that has been laid down in order to expedite travel or – for our purposes – trade execution. The form surrounds, envelopes, and guides a particular trade through legal hurdles to ultimately allow a proprietary trading strategy to deliver value;

2. **The "Wheels" – the ISDA Schedule:** reas the Pre-Print Form is standardized and seldomly (if ever) edited directly (the same way a highway is set in stone), the Schedule affords the parties the opportunity to elect and even negotiate provisions, such as adding in new sections which sit amicably with ongoing trends in the digital asset derivatives Industry. The Schedule is therefore the Wheels hitting the Highway – parties are permitted to choose the speed, style, and nature of their set of Wheels, but everyone is driving within the confines of the Highway;

3. **The "Fuel" – the Credit Support Annex (CSA):** The main purpose of a CSA is to provide for mechanisms that mitigate counterparty credit risk. It outlines the collateral arrangements between the parties and procedures for posting, valuing, and returning collateral. Certain key highlights of the CSA include;

   a) Collateral Eligibility and Valuation: This section specifies the classes of collateral that can be posted, whether be it traditional, fiat-based instruments or digital assets, as well as their applicable haircuts. It also outlines thresholds and other minimum amount to help mitigate ping-ponging collateral back and forth between counterparties;

   b) Posting and Withdrawing Collateral: This section outlines the procedures for posting and withdrawing collateral, including the frequency of collateral calls and the deadlines for posting or withdrawing collateral.

The CSA therefore allows the transaction to move forward by satisfying regulatory needs and mutually risk protection requirements.

4. **The "Driver" – the Confirmations:** SDA Confirmations are where SDM's key skills really shine through – it incorporates the vast majority of economic variables, such as the strike price and notional amount, of each Transaction. Although confirmations do often include legal terms as well, many parties opt to place these into an overarching "Master Confirmation Agreement" so that each party can focus on trading.

Thus, the major advantage of using an ISDA Master Agreement is that it standardizes the framework for derivative transactions, reducing the cost and complexity of negotiating individual contracts for each individual trade. It also provides a degree of legal certainty and protection for both parties in the event of a dispute.

It's important to note that ISDA agreements are complex legal documents, and should be reviewed and negotiated by experienced legal counsel. It is not a legal absolutism – it remains important to comply with all laws and regulations that have jurisdiction over a trade. Some areas of law, such as (but not limited to) bankruptcy, may have an impact on an ISDA's Netting Opinion.



CSA

A Credit Support Annex (CSA) is a legal document that is typically included as part of an ISDA (International Swaps and Derivatives Association) agreement. It outlines the collateral arrangements between the parties involved in a derivatives transaction. The main purpose of a CSA is to provide a mechanism for the protection of both parties in the event of a counterparty default.

A CSA generally includes several key components:

1. ollateral Requirements: This section specifies the types of collateral that can be posted (e.g., cash, government bonds, etc.) and the minimum credit quality requirements for the collateral.

2. Posting and Withdrawing Collateral: This section outlines the procedures for posting and withdrawing collateral, including the frequency of collateral calls and the deadlines for posting or withdrawing collateral.

3. Valuation of Collateral: This section specifies how the collateral will be valued, including the use of independent pricing services and the methods for resolving disputes over valuations.

4. Rehypothecation: This section outlines the rights and obligations of both parties regarding the re-use of collateral by the collateral taker.

5. Close-out Netting: This section explains the provisions for close-out netting which is triggered when one of the parties defaults. This allows the non-defaulting party to offset the outstanding obligations in order to limit the loss.

6. Termination Events: This section specifies the events that would trigger termination of the CSA, such as credit rating downgrade or bankruptcy.

7. Governing Law and Jurisdiction: This section explains the governing law and jurisdiction of the CSA.

The CSA is an important part of the ISDA agreement as it establishes the collateral arrangements between the parties, and it helps mitigate the counterparty risk in derivatives transactions. The CSA ensures that each party has access to sufficient collateral to cover the potential loss in case of a counterparty default. It's important to note that CSAs are complex legal documents and should be reviewed and negotiated by experienced legal counsel.

WHY WORK WITH SDM?

Started in 2017, we have completed over Billions in spot and derivatives transactions, and facilitated over $500M+ in loans in digital assets. SDM provides clients with the best rates through smart order routing across 20+ liquidity providers and the firm is connected to 7+ crypto native banks servicing clients across 60+ countries.

Specifically focused on digital assets, SDM provides a custom-fit approach to digital currency markets, with dedicated sales and bespoke trading to offer state-of-the-art trading solutions. Secure Digital Markets bridges the gap between institutional capital and digital assets by providing flexible and curated liquidity solutions to clients.

SDM and affiliated companies offer a suite of professional services to cover our client's immediate liquidity needs. These include:

Trading & Liquidity

- SDM's has a premier OTC desk and spot trading service for hedge funds, brokerages, treasuries, family offices, and miners to access best-in-class execution, pricing, and liquidity
- API Connectivity for live streaming and executable digital asset trading
- Fiat and Stablecoin on & off-ramp services for all of the major listed currencies

Derivatives

- Provide custom solutions for our clients' needs
- Excellent liquidity with consistent tight pricing and execution

Lending

- We provide access to non-recourse lending against the top 50 digital assets.

SDM also prides itself on maintaining a strong track record of reliable and secure service. Client assets and information are safe with us, and our compliance team makes sure that we are always adherent to the latest regulations.

SDM believes that by providing clients with everything needed in one place, our firm can help clients achieve their financial goals and make the most of the exciting opportunities presented by the digital asset market.